GCI World 2026 September

In-Class Competition Tutorial

松尾・岩澤研究室
Matsuo-Iwasawa Lab UTokyo

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

Agenda

- What is In-Class Competition?
- How to Submit

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

(Recap) Course Completion Requirements

You will be evaluated on the following assignments:

	Total	Completed	Honors	Outstanding
 Attendance Surveys	14	7+	7+	Selected Students (based on overall scores)
 Homework	24 points (=8 HW) *3 points max, 2 points if late	14+ points	14+ points	
 Final	1	Pass	TOP 10% AND TOP 20%	
 Competition	1	(optional)		

You need to participate in the competition to become Honors/Outstanding Student.

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

What is In-Class Competition?

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

About In-Class Competition

Students compete with one another based on their scores through In-Class Competition

Do your best to achieve a high ranking!!!

*During the competition, public leaderboard will be visible. After the deadline, we will conduct the final evaluation on a separate private leaderboard.

Leaderboard

RANK	NAME	SCORE
🏆	25UT_Luffy	0.74838
2	25UT_Naruto	0.74114
3	25UT_Tanjiro	0.73355

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

About In-Class Competition

The in-class competition is modified version of the **“Home Credit Default Risk”** competition originally held on Kaggle.

Your task is to **build a model to predict the probability of customer loan default based on the provided customer data.**



Source: Kaggle, “Home Credit Default Risk” — Image: Home Credit Group
https://www.kaggle.com/competitions/home-credit-default-risk/overview/description

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

Submission Deadline

Deadline: November 20th 1:00 AM UTC

Submissions open: October 2nd 11:00 AM UTC

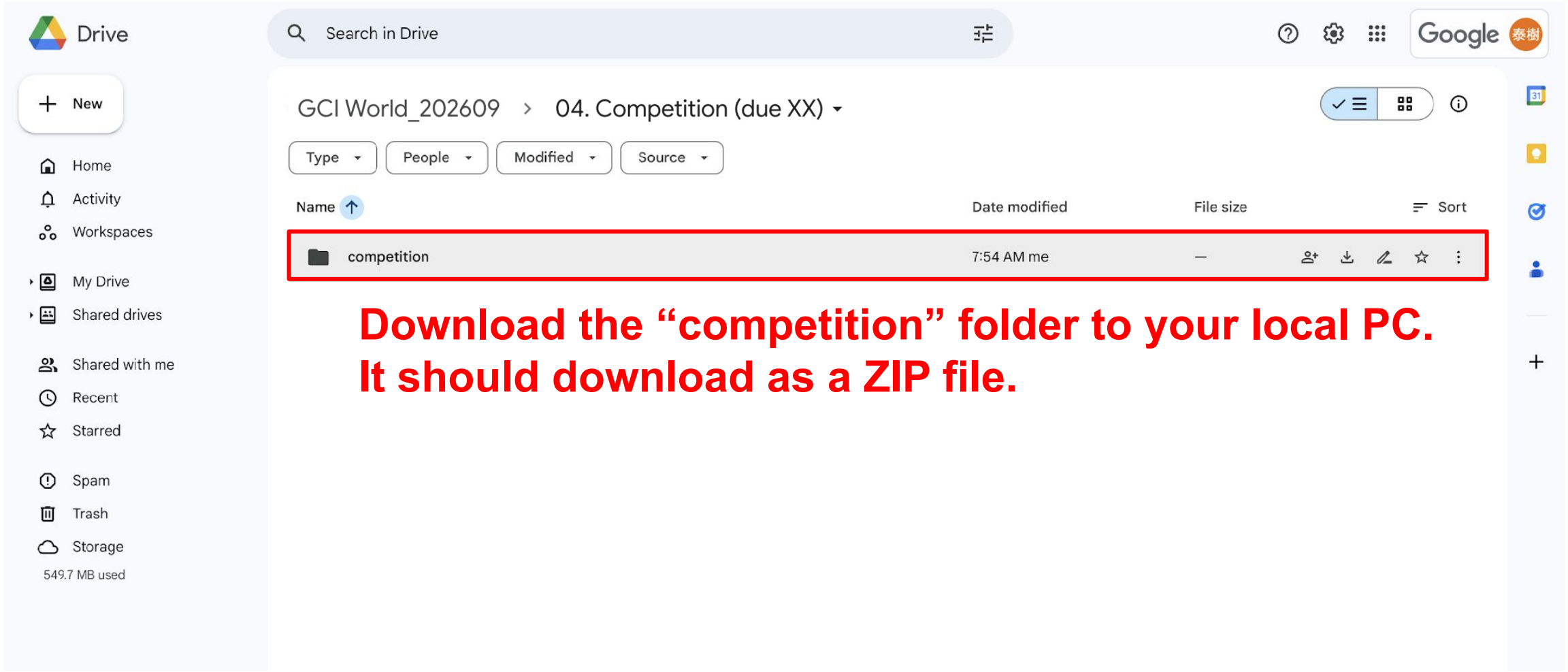
© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

How to Get Started

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

1. Download the materials

Download the “competition” folder from the class materials



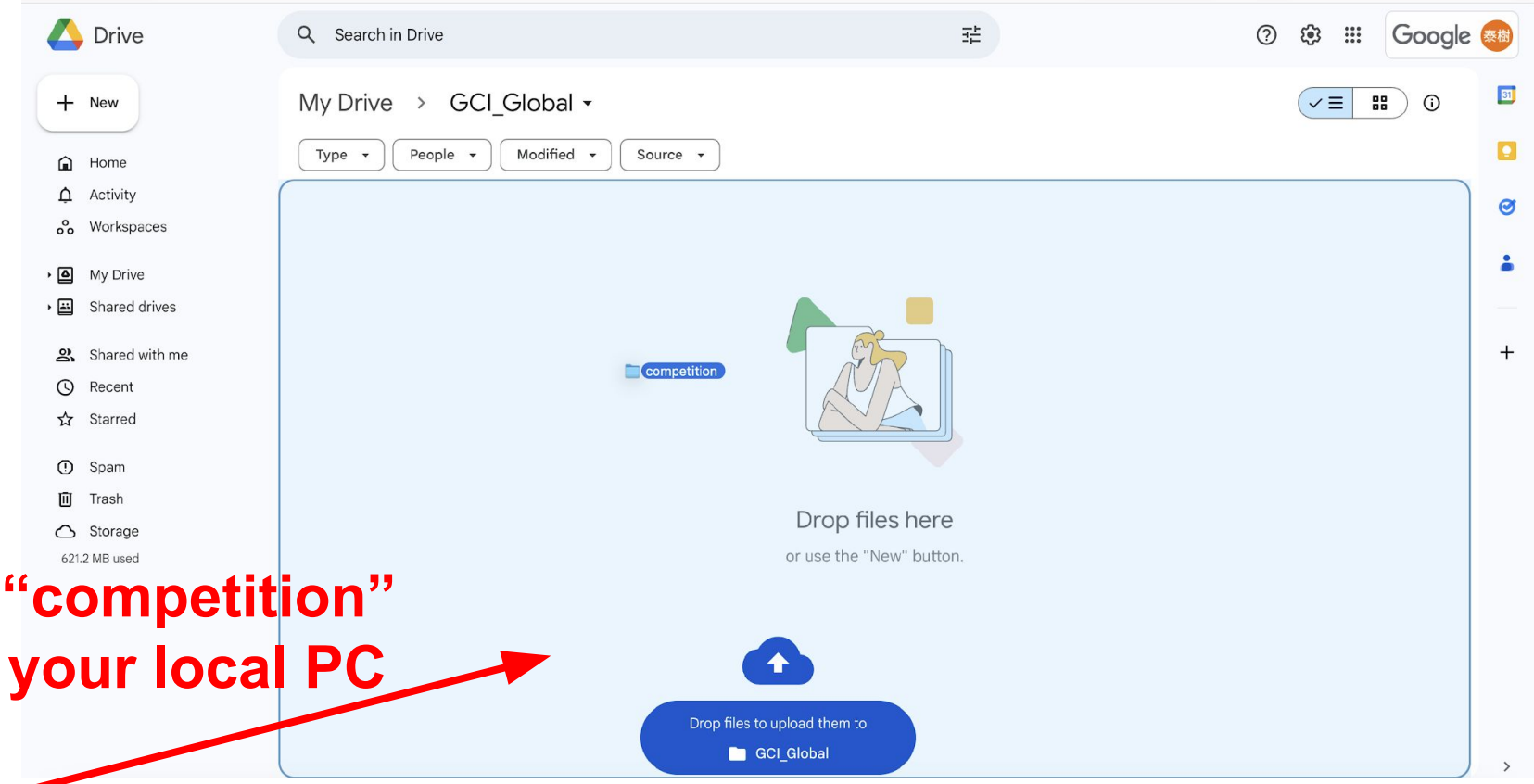
© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

2. Upload the materials to your Google Drive

Unzip the downloaded ZIP file.

Then, upload the “competition” folder to your MyDrive.

(It can be anywhere in your MyDrive, but you will need to adjust the PATH accordingly later)

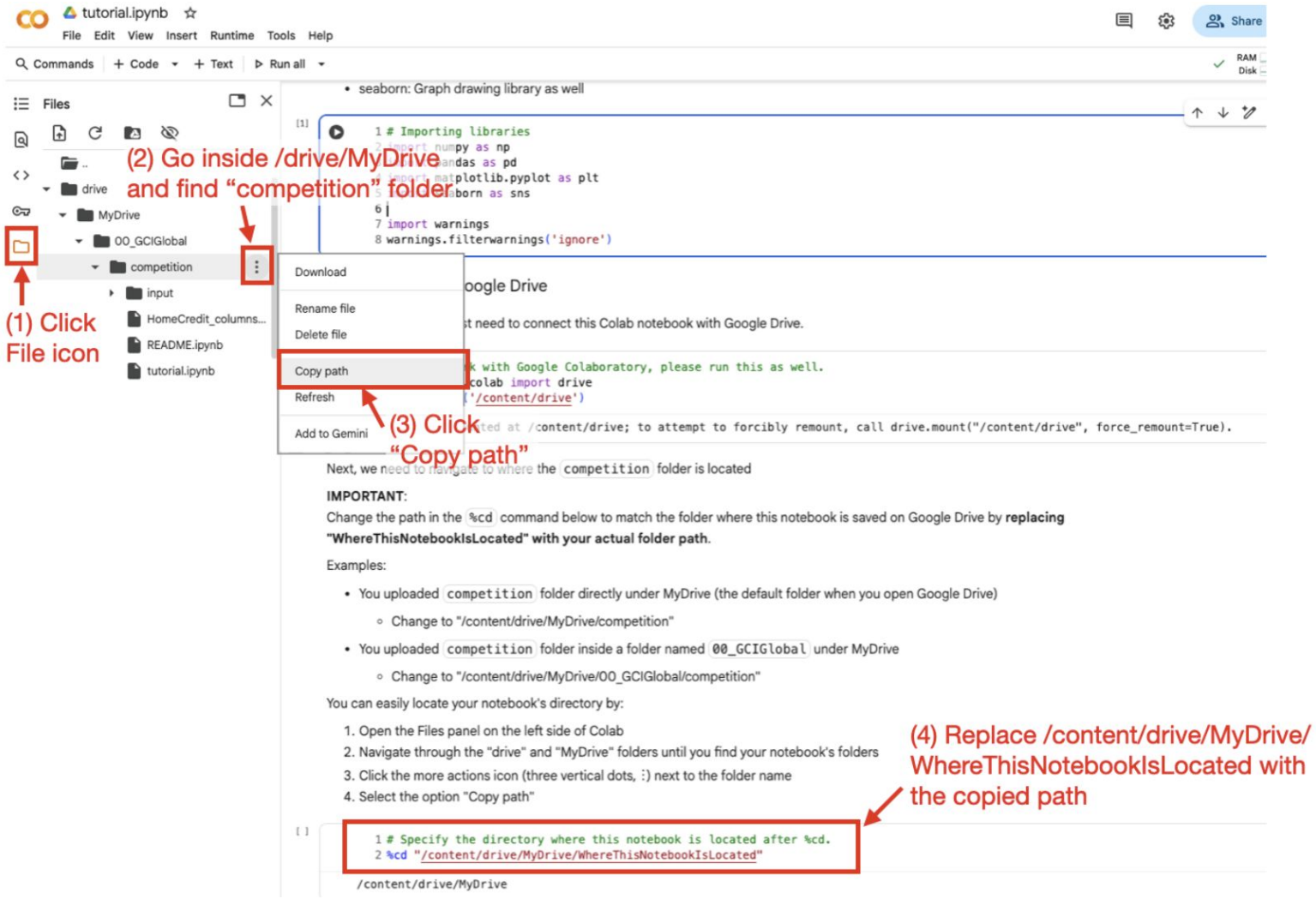


Upload the “competition” folder from your local PC

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

3. Open the tutorial.ipynb inside “competition” folder

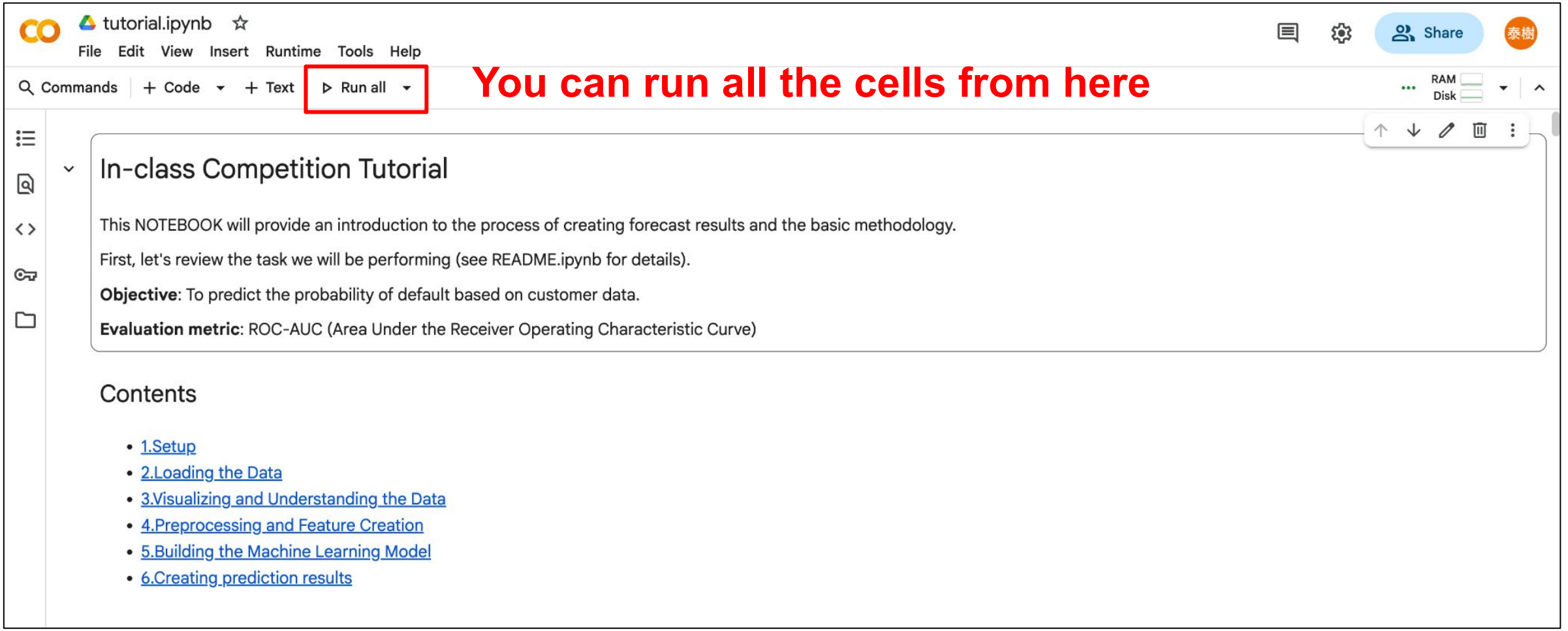
Scroll down a bit and change the PATH so that it points to the directory where you uploaded the “competition” folder



© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

4. Run the tutorial to create submission.csv

Please run all the cells in“tutorial.ipynb” to create “submission.csv”

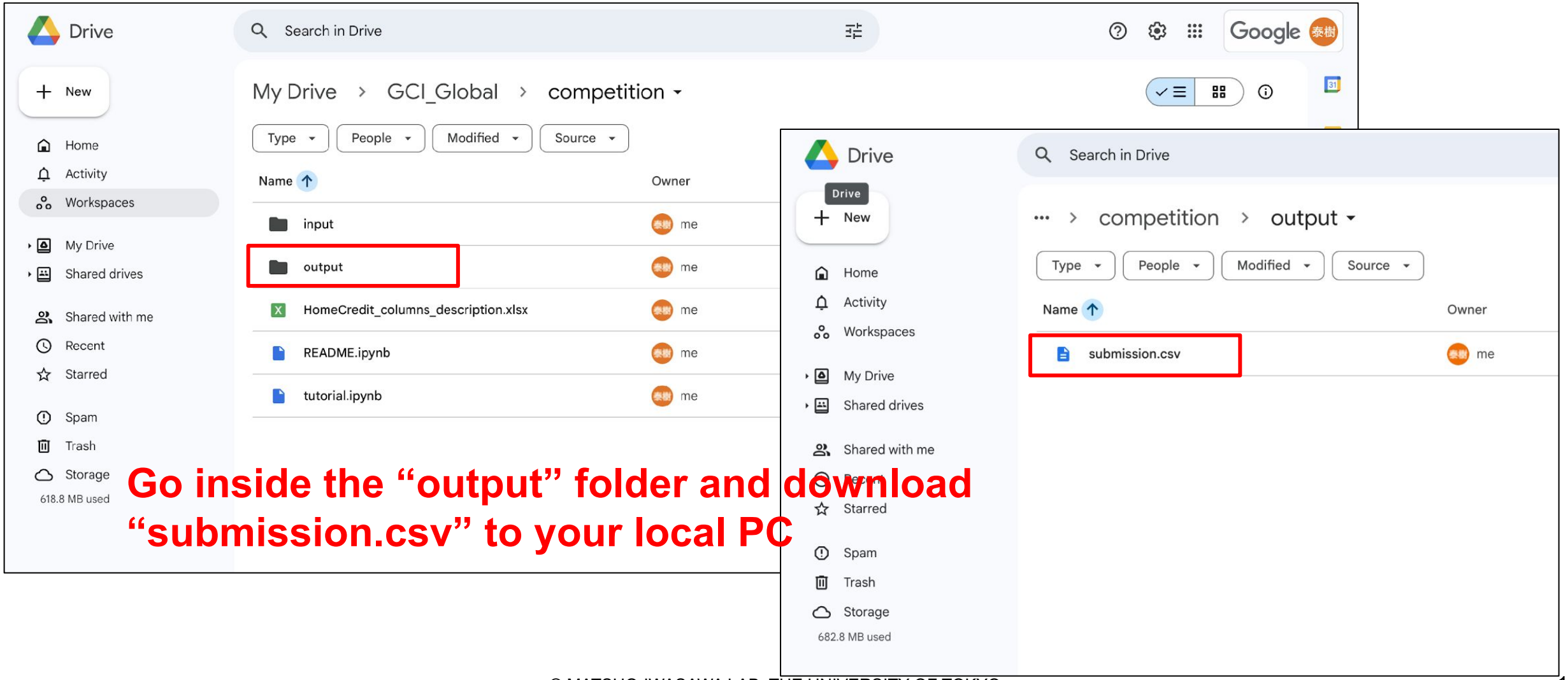


Run “tutorial.ipynb” to create “submission.csv”

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

5. Download the submission.csv

You will find a new folder named “output”. Download the “submission.csv” inside the folder.



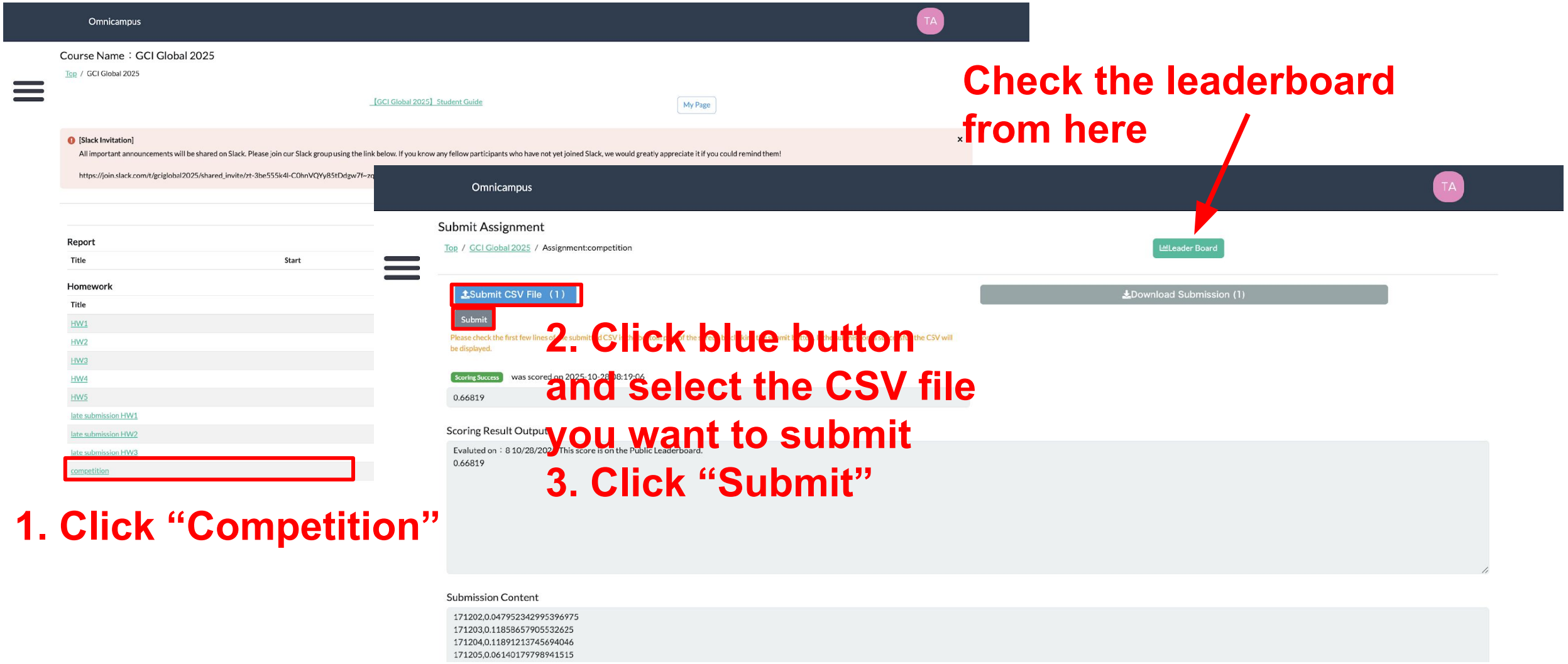
© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

How to Submit

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

Submit on Omnicampus

Login to Omnicampus to submit your prediction and see the score.



© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

How to Submit Your Code (if aiming for Honors/Outstanding)

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

Submitting Your Code (e.g. Colab)

If you are aiming to get selected as Honors/Outstanding Students, you **MUST** submit your code as well.

e.g. if you are writing your code using “tutorial.ipynb”.... 👇

6.3 Saving this notebook as a ZIP file [REQUIREMENT IF AIMING FOR HONORS/OUTSTANDING STUDENTS]

This section compresses the notebook file (`tutorial.ipynb`) into a ZIP file and saves it to your working directory. Make sure notebook (Ctrl+S or Cmd+S) before running this cell to include the latest changes.

If you are aiming to become **Honors or Outstanding Student**, you need to submit the ZIP file as well on Omnicampus.

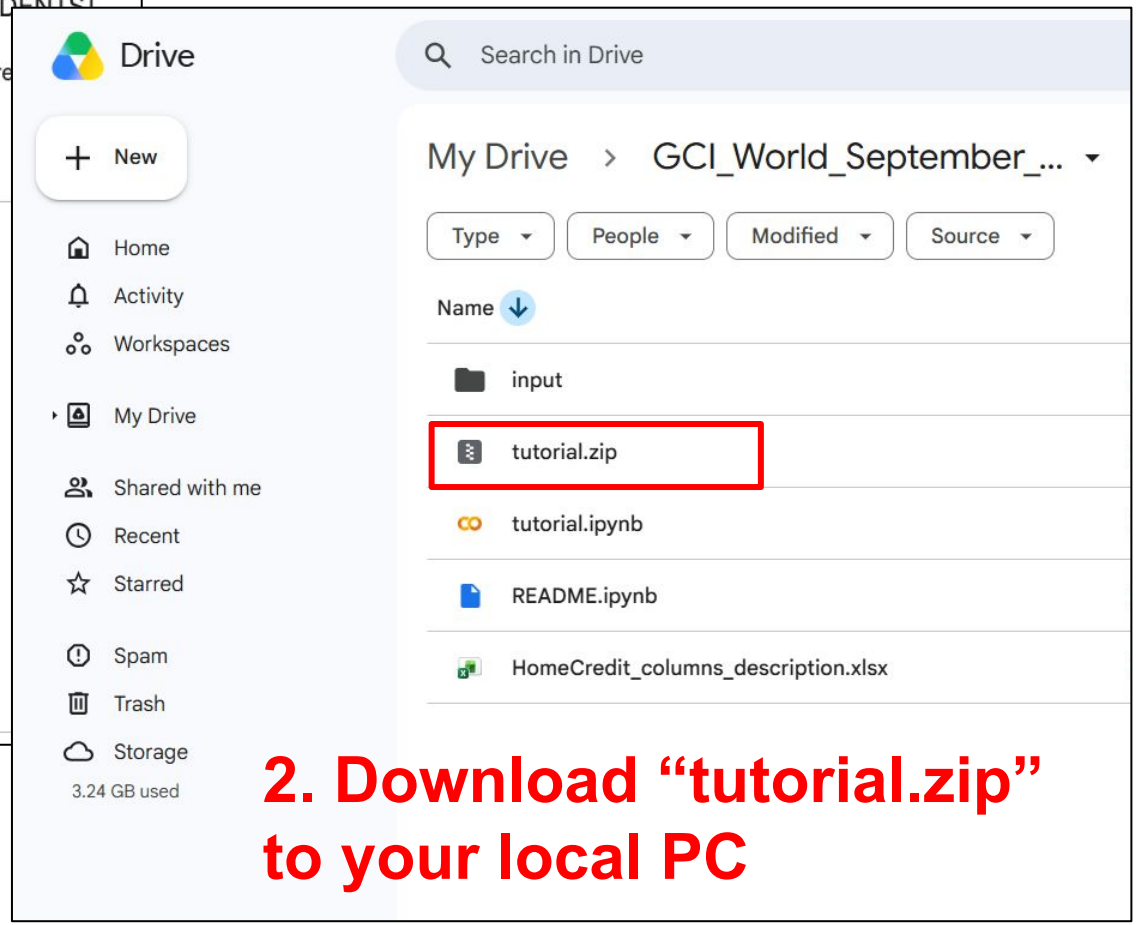
```python
import zipfile

# Define the file names
notebook_filename = "tutorial.ipynb" # Enter the name of this notebook
zip_filename = "tutorial.zip"

# Define the full paths using current_dir
notebook_path = current_dir / notebook_filename
zip_output_path = current_dir / zip_filename

# Check if the notebook file exists and create the ZIP file
if notebook_path.exists():
    with zipfile.ZipFile(zip_output_path, 'w', zipfile.ZIP_DEFLATED) as zipf:
        # Add the notebook to the ZIP file (arcname avoids including the full folder structure)
        zipf.write(notebook_path, arcname=notebook_filename)
    print(f"Successfully created: {zip_output_path}")
else:
    print(f"Error: Could not find '{notebook_filename}' in {current_dir}.")
    print("Please check if the notebook name and the current directory are correct.")
```

1. Run the last cell in “tutorial.ipynb”

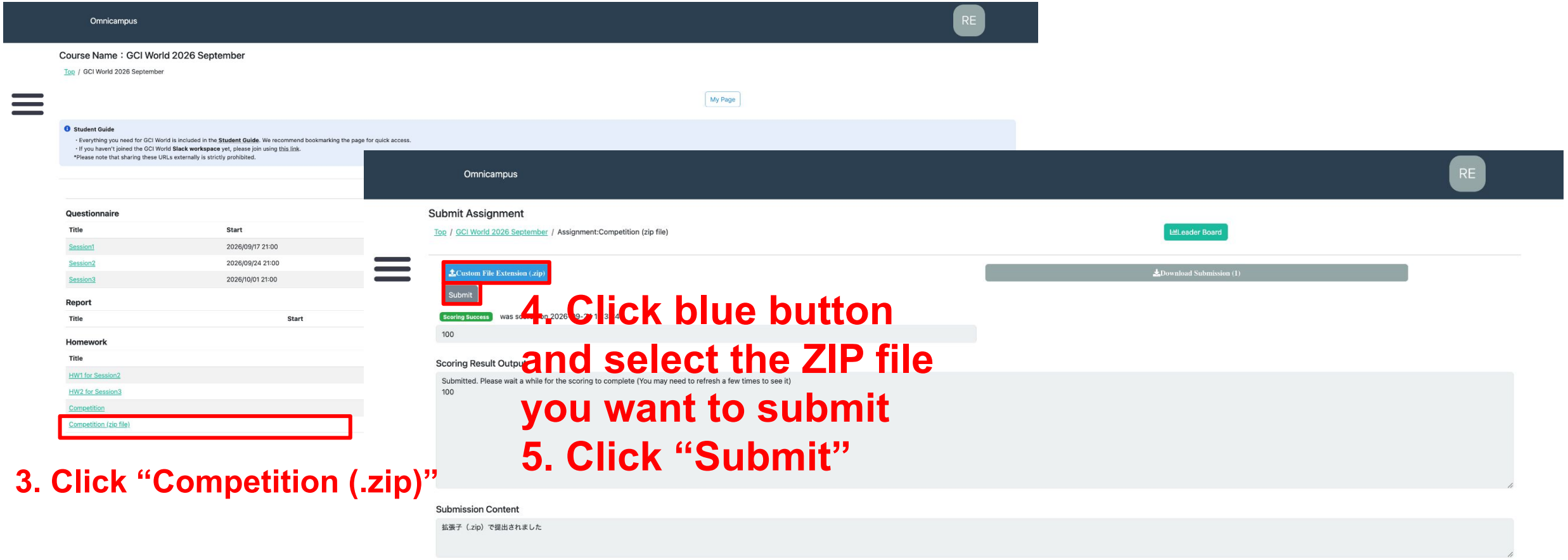


2. Download “tutorial.zip” to your local PC

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

Submitting Your Code (e.g. Colab)

If you are aiming to get selected as Honors/Outstanding Students, you **MUST** submit your code as well.



© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

Submission Deadline

Deadline: November 20th 1:00 AM UTC

Submissions open: October 2nd 11:00 AM UTC

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

Next Steps

You can use the “tutorial.ipynb” to customize your model.

Read the README.ipynb in “competition” folder for the details and rules of the competition.

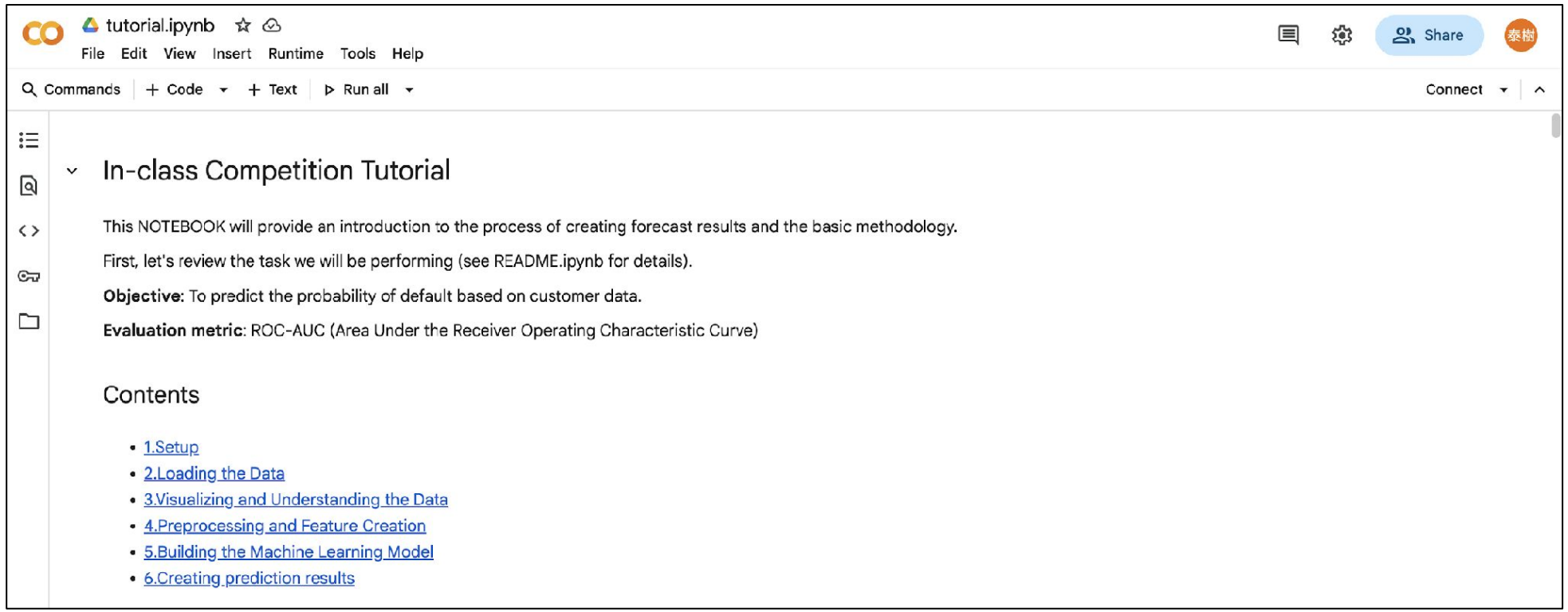
Take a look at “HomeCredit_columns_descriptions.xlsx” for details about the dataset.

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

Note: About the tutorial.ipynb

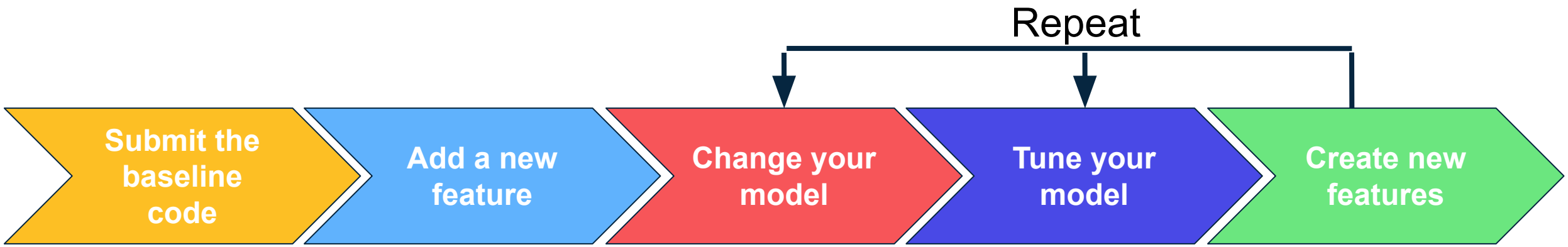
The tutorial.ipynb notebook is a simple baseline designed for beginners.
By running it, you can generate a CSV file ready for submission to Omnicampus.

You can improve the score by **adding more features, changing the model, and tuning the hyper-parameters**.



© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

Competition Workflow (How to improve your model)



Step 1
Just run the baseline, and you can **submit the baseline code to join the competition**!

Step 2
Add more features. Use visualizations to understand the data better.

Step 3
Try a new model, like **AdaBoost, XGBoost, CatBoost or LightGBM**, or *an ensemble of them (advanced)*.

Step 4
Tune the hyper-parameters of the model using **grid search** or *Optuna (advanced)*.

Step 5
Create new features from existing ones. It can be something meaningful, or *all possible cross features (advanced)*.

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO

松尾・岩澤研究室
MATSUO-IWASAWA LAB UTOKYO

© MATSUO-IWASAWA LAB, THE UNIVERSITY OF TOKYO